THE INVARIANTS OF THE INERTIA GROUP FORM THE INTEGRAL CLOSURE OF A GALOIS SUBEXTENSION, AND THE CORRESPONDING LOCAL EXTENSION IS ETALE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'M_1 = B^I is the integral closure' inside the counted proof span. The nine references inside the counted proof span are recorded as non-load-bearing navigation pointers; they include cross-chapter tags and same-chapter label lines.

Allowed standard prerequisites (editorial).

- The inertia group of a maximal ideal in a Galois extension of a discretely valued field, and the decomposition group. fixed at statement lines 33477-33479 and used throughout the counted proof span. Supplies the group I and the quotient D/I whose action on the residue field is analysed.
- The norm of a uniformizer and multiplicativity of ramification indices in a tower of discretely valued fields. used at proof lines 33503-33520. Produces the two degree formulae whose comparison yields the etaleness conclusion.
- Purely inseparable residue extension for the invariants subgroup, and faithfulness of the action of D/I. used at proof lines 33490-33498. Gives the bound [kappa(m') : kappa] >= |D/I| that the index comparison consumes.

Source reference. Original lemma 00U6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1100–1100. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3446–3446. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 09I1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12024–12024. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09I3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12026–12026. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 33179: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33179–33179. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32553: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32553–32553. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32680: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32680–32680. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32704: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32704–32704. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33006: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33006–33006. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K. Let L/K be a finite Galois extension. Let $\mathfrak{m} \subset B$ be a maximal ideal of the integral closure of A in L. Let $I \subset G$ be the inertia group of $\mathfrak{m}$. Then B^I is the integral closure of A in L^I and $A \to (B^I)_{B^I \cap \mathfrak{m}}$ is étale.*

Proof. Write $B' = B^I$. It follows from the definitions that $B' = B^I$ is the integral closure of A in L^I. Write $\mathfrak{m}' = B^I \cap \mathfrak{m} = B' \cap \mathfrak{m} \subset B'$. By Lemma 32680 the maximal ideal $\mathfrak{m}$ is the unique prime ideal of B lying over $\mathfrak{m}'$. As I acts trivially on $\kappa(\mathfrak{m})$ we see from Lemma 32553 that the extension $\kappa(\mathfrak{m})/\kappa(\mathfrak{m}')$ is purely inseparable (perhaps an easier alternative is to apply the result of Lemma 32704). Since D/I acts faithfully on $\kappa(\mathfrak{m}')$, we conclude that D/I acts faithfully on $\kappa(\mathfrak{m})$. Of course the elements of the residue field κ of A are fixed by this action. By Galois theory we see that $[\kappa(\mathfrak{m}') : \kappa] \geq |D/I|$, see Fields, Lemma 09I3.

Let π be the uniformizer of A. Since $\mathrm{Norm}_{L/K}(\pi) = \pi^{[L:K]}$ we see from Algebra, Lemma 02MJ that

$$|G| = [L : K] = [L : K] \, \mathrm{ord}_A(\pi) = |G/D| \, [\kappa(\mathfrak{m}) : \kappa] \, \mathrm{ord}_{B_\mathfrak{m}}(\pi)$$

as there are $n = |G/D|$ maximal ideals of B which are all conjugate under G, see Remark 33006 and Lemma 33179. Applying the same reasoning to the finite extension the finite extension L/L^I of degree $|I|$ we find

$$|I| \, \mathrm{ord}_{B'_{\mathfrak{m}'}}(\pi) = [\kappa(\mathfrak{m}) : \kappa(\mathfrak{m}')] \, \mathrm{ord}_{B_\mathfrak{m}}(\pi)$$

We conclude that

$$\mathrm{ord}_{B'_{\mathfrak{m}'}}(\pi) = \frac{|D/I|}{[\kappa(\mathfrak{m}') : \kappa]}$$

Since the left hand side is a positive integer and since the right hand side is ≤ 1 by the above, we conclude that we have equality, $\mathrm{ord}_{B'_{\mathfrak{m}'}}(\pi) = 1$ and $\kappa(\mathfrak{m}')/\kappa$ has

degree $|D/I|$. Thus $\pi B'_{\mathfrak{m}'} = \mathfrak{m}' B'_{\mathfrak{m}}$ and $\kappa(\mathfrak{m}')$ is Galois over κ with Galois group D/I, in particular separable, see Fields, Lemma 09I1. By Algebra, Lemma 00U6 we find that $A \to B'_{\mathfrak{m}'}$ is étale as desired. $\square$